

Active Management of IO Beta

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IO Beta Allocation Strategy: January 2016

Summary

- ▶ **Actual Allocation:** 17%
- ▶ **Model 1 (IO Px/Cpn not duration-adjusted):** 18%.
(December Prediction: 20%)
- ▶ **Model 2 (IO Px/Cpn duration-adjusted):** 10%.
(December Prediction: 6%)

Active Management of IO Beta

Investigate if we can design an **active** strategy for investing in IO Beta that outperforms a **passive** strategy that is always 100% invested.

Metrics and Set-up

- ▶ Create a **Hedged IO** index
 - ▶ Hedge various IOS with current coupon TBA using model durations (Barclays)¹
 - ▶ Index is created by using monthly market value weights for each IOS based on the market weight of the corresponding cohort
 - ▶ Ginnie Mae IOS are excluded
- ▶ Simulating Passive IO Beta: Total Return and Sharpe Ratio of Hedged IO index from March 2011 - December 2015
- ▶ Simulating Active IO Beta: Dynamic allocation to Hedged IO index based on relative value metrics

¹Note that we do *not* hedge out the curve risk and volatility risk associated with an individual IOS.

Performance of Passive IO Beta Strategy: Mar 2011 - Dec 2015

- ▶ **CAGR:** 9.8%
- ▶ **Ann. Volatility:** 16.5%
- ▶ **Sharpe Ratio:** 0.7
- ▶ **% Positive Months:** 60
- ▶ **Best Month:** 11.6%
- ▶ **Worst Month:** -12.9%
- ▶ **Max Drawdown:** -23.5%
- ▶ **Drawdown Period:** June 2011 - March 2013

Framework for Active IO Beta Strategy

- ▶ Regress Hedged IO Index returns against RV metrics (1m lag):
 - ▶ IO Index Price/Coupon²
 - ▶ Projected 1-month Carry³
 - ▶ Convexity: 60-day moving average of realized volatility on the current coupon
 - ▶ Prepayment Risk: Incentive for the Mortgage Universe
 - ▶ Yield Curve Slope: 10s-2s slope
- ▶ Calculate z-score of projected return and allocate to Beta (Net IO) based on score

²While OAS would also be useful, in practice dealer model updates are not always appropriately back-filled which makes it challenging to get a consistent time series of values.

³This is not available on a historical basis so we use realized carry as a proxy.

Active IO Beta Strategy: Regression Results

- ▶ Significant Variables
 - ▶ High: IO Index Price/Coupon, Prepayment Risk
 - ▶ Medium: Realized Volatility
 - ▶ Low: Realized Carry, Yield Curve Slope, Index OAS
- ▶ Final Model
 - ▶ Proj. Return $\sim P_x/C_{pn} + C_{nvx} + \text{PrepayRisk} + \text{Carry}$

Active IO Beta Strategy: Beta Allocation

- ▶ Solve for $k = a + bz$ ($z = z\text{-score}$) such that we optimize the Sharpe Ratio of the strategy $k \times \text{Hedged IO Return}$.
- ▶ Set $\text{IO Beta}_{\max} = 100\%$ and $\text{IO Beta}_{\min} = -0.25\%$
- ▶ Rescale k so that it is (more or less) mapped to these boundaries.

Active IO Beta Strategy: Two Beta Allocation Models

- ▶ Model 1: Don't adjust IO Index P_X/C_{pn} for Interest Rates. $R^2 = 0.29$.
- ▶ Model 2: Adjust IO Index P_X/C_{pn} for Interest Rates.⁴ $R^2 = 0.20$.

⁴We estimate that the P_X/C_{pn} ratio moves by about 0.14 for each 10bps move in the current coupon.

Active IO Beta Strategy: Model 1 Return Profile

- ▶ **CAGR:** 9.9%
- ▶ **Ann. Volatility:** 6.0%
- ▶ **Sharpe Ratio:** 1.6
- ▶ **% Positive Months:** 74
- ▶ **Best Month:** 9.8
- ▶ **Worst Month:** -1.0%
- ▶ **Max Drawdown:** -1.5%
- ▶ **Drawdown Period:** November 2011 - March 2012

Active IO Beta Strategy: Model 2 Return Profile

- ▶ **CAGR:** 7.9%
- ▶ **Ann. Volatility:** 6.3%
- ▶ **Sharpe Ratio:** 1.2
- ▶ **% Positive Months:** 72
- ▶ **Best Month:** 9.6
- ▶ **Worst Month:** -2.9%
- ▶ **Max Drawdown:** -3.7%
- ▶ **Drawdown Period:** November 2011 - March 2012

Track Record of IO Beta Allocation Strategies

	IO Beta Allocation			Monthly Hedged Return						Yearly Return				
	Fund	Model1	Model2	100% Beta	100% Beta CrvAdj	Fund	Model1	Model2	100% Beta	100% Beta CrvAdj	Fund	Model1	Model2	
Jan-15	26%	-8%	-10%	-8.8%	-8.9%	-2.31%	0.71%	0.89%						
Feb-15	25%	43%	66%	8.2%	5.5%	1.38%	2.37%	3.63%						
Mar-15	21%	-3%	-44%	-0.4%	1.2%	0.25%	-0.04%	-0.53%						
Apr-15	15%	-18%	-12%	4.5%	1.2%	0.18%	-0.22%	-0.14%						
May-15	22%	-22%	-25%	0.1%	-1.7%	-0.37%	0.37%	0.43%						
Jun-15	31%	-10%	-13%	1.6%	-0.7%	-0.20%	0.07%	0.09%						
Jul-15	33%	-20%	-41%	-0.1%	1.2%	0.40%	-0.24%	-0.49%						
Aug-15	28%	-30%	-13%	-0.3%	-1.5%	-0.42%	0.45%	0.20%						
Sep-15	24%	-41%	-25%	-0.5%	-1.6%	-0.38%	0.66%	0.40%						
Oct-15	26%	-2%	12%	-0.3%	-0.1%	-0.03%	0.00%	-0.02%						
Nov-15	21%	5%	0%	1.5%	1.8%	0.38%	0.09%	0.00%						
Dec-15	22%	20%	6%	0.1%	1.4%	0.30%	0.27%	0.08%	5.61%	-2.23%	-0.85%	4.50%	4.53%	
Jan-16	17%	18%	10%											

As of: Jan 5, 2016

Evolution of IO Beta Allocation RV Variables

Date	MthlyHedgedReturn	OAS	Carry	RealizedVol	YCSlope	PrepayRiskPrem	PxCpnMult	LIB1m	RefiPctPool	RefiPctLoan	RefiIndex	CC	CCChanges	CurveHedgedReturn
1/30/2015	-11.821	65	1.1038	60.22	120.8	102.97	4.78	0.1713	0.6626	0.6558	2585	2.45	-38.12	-8.9368
2/27/2015	11.1577	-87	1.6224	73.25	137.6	88.62	5.51	0.173	0.611	0.5614	1740	2.74	28.39	5.5289
3/31/2015	-0.7	-97	0.9897	78.46	137.9	92.38	5.37	0.1763	0.642	0.6199	1942	2.66	-7.56	1.1795
4/30/2015	5.207	28	0.2384	69.09	146.7	80.98	5.56	0.181	0.6364	0.6089	1559	2.75	8.7	1.1912
5/29/2015	1.2501	27	0.3672	68.4	149.2	72.75	5.66	0.184	0.5495	0.4952	1234	2.81	6.45	-1.6641
6/30/2015	4.679	10	0.3215	75.45	169.8	50.56	6.16	0.1865	0.45	0.397	1246	3.07	26.07	-0.6597
7/31/2015	-1.5451	14	0.1851	82.06	153.1	66.46	5.93	0.1918	0.4991	0.4227	1418	2.92	-15.78	1.1809
8/31/2015	0.1421	21	0.4859	76.13	146.2	67.72	5.92	0.1986	0.5478	0.5324	1546	2.93	1.89	-1.5283
9/30/2015	-1.2489	40	0.6095	67.08	141.5	70.97	5.7	0.193	0.5189	0.4886	1590	2.8	-13.25	-1.5878
10/30/2015	0.1931	42	0.5574	66.44	141.6	68.26	5.69	0.192	0.5789	0.5526	1535	2.84	4.25	-0.1309
11/30/2015	2.3941	3	0.5458	57.52	128.7	57.14	5.89	0.243	0.4759	0.4481	1489	2.95	10.3	1.7776
12/31/2015	1.1442	5	0.5949	61	121.7	52.9	5.97	0.4295	0.4466	0.4113	1667	3.01	5.79	1.36

Allocation Grid for IO Basis

	PxCpnMult					
PrepayRiskPrem	5.72	5.82	5.92	6.02	6.12	6.22
40.4	69	59	49	39	30	20
45.4	58	48	39	29	19	9
50.4	48	38	28	19	9	-1
55.4	37	28	18	8	-2	-11
60.4	27	17	7	-2	-12	-22
65.4	16	7	-3	-13	-22	-32